

Most PCs today—i.e., if you own a processor like the AMD Athlon (90/65nm) or Core 2 Duo and above—come with decent headroom to increase performance. The question is: have you ever considered pushing the limits?

If so, tighten your seat belt, because in this article we will not only learn about torturing the hardware, but also unravel the facts about overclocking your hardware and the status of benchmarking in Linux.



Note: Overclocking is raising the clock speed of the processor from the factory settings. Benchmarking, on the other hand, is an assessment test to measure the capacity of the hardware at your disposal.

Some of you must be mulling over what Linux has to do with overclocking? Well, yes, the operating system has very little to do with overclocking. But when you pump up the speed of your system, then various things need to be considered. The operating system chips in by offering various monitoring tools and other requisites.

Note that this tutorial will only guide you with the basics of overclocking. Since it's a very vast topic, and the overclocking options vary from motherboard to motherboard and the BIOSes they come with, more details are out of scope of this article.



Statutory warning: Overclocking is fun to begin with, but reckless decisions can bring your whole system down. The author and the publisher will not be responsible for any damage caused, and what's worse is that afterwards, you will not be covered by the product warranty either. So, you'll need to remember one thing—slow and steady wins the race.

The only precautionary measure is not to take hasty decisions—stay calm, and nothing will go wrong. The fact is, overclocking is highly addictive, so beware and don't let your excitement overtake you. Because once you get overexcited, you will surely end up doing something wrong.

Now that we're done with the warnings and introductory notes, let's move on. Overclocking requires a few tools and utilities as well. But for what, you may ask? As I mentioned earlier, overclocking needs to be done through the BIOS. Yes, the best and rudimentary way to overclock is from BIOS. However, once you have overclocked the system, there are several factors that need to be dealt with.

The most important is the temperature. Temperature plays a vital role in the life of silicon. All the diodes/chips/processors are made using silicon, so the temperature must be kept under control. We will get to know more about it later. But before that we need some software to monitor certain things for us.

Unfortunately, the software potpourri isn't as diverse as that of Windows, but still, there is a decent amount of benchmarking and monitoring software available for Linux. In this article we will take a look at the following:

1. **Hardinfo:** A one-stop shop for all the monitoring utilities in Linux
2. **Phoronix Text Suite:** Undoubtedly, a vast and the most popular benchmarking suite for Linux
3. **Geekbench:** Another cross-platform benching program
4. **Prime95B:** A CPU stress-testing program

The test set-up used for overclocking and benchmarking is described below:

- AMD Phenom II Dual Core X2 550 Black Edition processor (unlocked) to Phenom II Quad Core X4 B50), with 45nm fabrication and socket AM3
- Jetway Ha07 790GX AM2+ motherboard
- Transcend DDR2 800 MHz 4GB kit @ 5-5-5-15/dual channel
- Western Digital 640AAKS Blue Edition hard disk drive
- Prolimatech Megahalems + Scythe Sflcx 120mm 75cfm fan (CPU heat sink and fan)
- Corsair VX450 SMPS | open air system
- Operating systems: Ubuntu 9.04 64-bit and Mandriva 2009.1 32-bit

Benchmarking your PC

Before we get started, we should know about the types of benchmarking. There are many benchmarking applications and methods to test the system. These are generally segregated into the following two categories:

1. **Real-world benchmarking:** All benchmarking of real tests such as FPS (frames per second) in games, the time to compile software/the kernel, the encoding/decoding time of media files, etc, fall under this category. This is perhaps a more practical way to judge a PC's calibre. Most benchmarking software don't cater to real-world benchmarking; thus, these tests are done by using the computer on a daily basis.
2. **Synthetic benchmarking:** All software and tests designed to appraise the system potential are termed as synthetic benchmarking software. These are good to obtain quick results, and one doesn't need to waste much time running tests, one by one. However, some of the software tends to favour certain brands and products. For example, the Super PI test favours Intel processors while the AES benchmarking favours AMD processors. Thus, these are not always the premier solution to judge a PC's potential.

Hardinfo

Sleuth tools let you obtain crucial information about your hardware. These tools not only provide valuable information but also prove to be viable alternatives to digital multimeters and power calculators. Unfortunately, there aren't many alternatives available on the Linux platform, and this is where Hardinfo chips in to provide the perfect